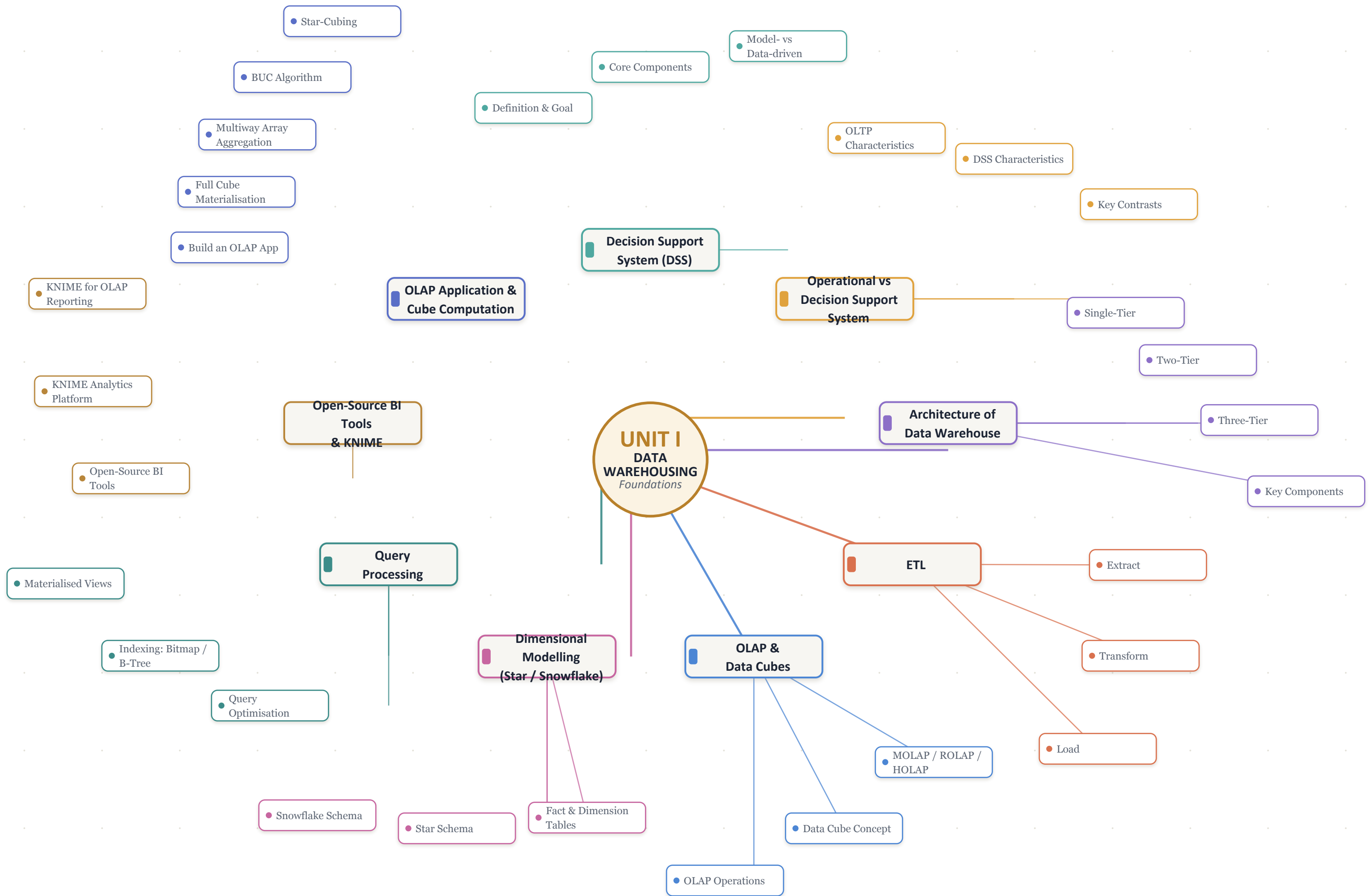


UNIT I

Introduction to Data Warehousing

A structured concept map covering Decision Support Systems, Warehouse Architecture, ETL, OLAP & Data Cubes, Dimensional Modelling, Query Processing, Open-Source BI Tools & KNIME, and Data Cube Computation Methods.

UNIT I — INTRODUCTION TO DATA WAREHOUSING · CONCEPT MAP



Decision Support System (DSS)

Decision Support Systems assist managers in making decisions by analysing large amounts of data. Unlike routine transaction systems, a DSS supports semi-structured and unstructured problem solving through interactive access to models and data.

- **Definition & Goal**

A DSS is a computer-based information system that combines data, analytical models and a user interface to help managers make semi-structured and unstructured decisions.

- **Core Components**

Three core components: the Data Management subsystem (databases), the Model Management subsystem (analytical models), and the interactive User Interface.

- **Model- vs Data-driven**

DSS types are classified as Model-driven (relies on financial / simulation models) or Data-driven (relies on large historical datasets, e.g. OLAP).

Operational vs Decision Support System

Operational databases run the business in real time, while Decision Support Systems analyse historical data drawn from many operational sources to support strategic and tactical decisions.

- **OLTP Characteristics**

Operational systems (OLTP) handle day-to-day transactions, store current, detailed, frequently updated data, and are optimised for fast insert / update / delete operations.

- **DSS Characteristics**

Decision-support (OLAP) systems store historical, summarised, subject-oriented data optimised for complex read-only analytical queries.

- **Key Contrasts**

Contrasts span Purpose (run the business vs analyse the business), Data (current vs historical), Design (normalised ER vs denormalised dimensional), and Users (clerks vs managers / analysts).

Architecture of Data Warehouse

The architecture of a data warehouse defines how data flows from operational sources through staging and integration into a central repository, and out to analytical tools, typically organised in one, two, or three tiers.

- **Single-Tier**

A minimal architecture that compresses source, staging and analysis layers to remove redundancy; rarely used in practice.

- **Two-Tier**

Separates source systems from the warehouse server, but shares one tier for staging and presentation - common in small deployments.

- **Three-Tier**

Standard architecture: Bottom tier (sources + staging + ETL), Middle tier (OLAP server), Top tier (front-end reporting / analysis tools).

- **Key Components**

Source Systems, Staging Area, the central Data Warehouse, dependent Data Marts, and the Metadata repository.

ETL

ETL (Extract, Transform, Load) is the pipeline that moves data from source systems into the data warehouse, ensuring it is clean, consistent and ready for analysis.

- **Extract**

Data is pulled from heterogeneous operational sources such as databases, flat files, ERP systems and web services.

- **Transform**

Extracted data is cleaned, de-duplicated, validated, standardised and aggregated into the warehouse's dimensional structure.

- **Load**

Transformed data is loaded using full loads (initial) or incremental loads (refresh), usually in scheduled batch windows.

OLAP & Data Cubes

On-Line Analytical Processing (OLAP) enables fast, interactive analysis of multidimensional data organised as data cubes, supporting operations like roll-up, drill-down, slice, dice and pivot.

- **MOLAP / ROLAP / HOLAP**

MOLAP stores pre-aggregated cubes in multidimensional arrays; **ROLAP** queries relational tables directly; **HOLAP** combines both.

- **Data Cube Concept**

A data cube organises data by multiple dimensions (e.g. Time, Product, Location) with measures (e.g. Sales) at each cell, enabling multi-angle analysis.

- **OLAP Operations**

Roll-up (aggregate up a hierarchy), **Drill-down** (finer detail), **Slice** (fix one dimension), **Dice** (sub-cube from 2+ dims), **Pivot** (rotate axes).

Dimensional Modelling (Star / Snowflake)

Dimensional data modelling structures a warehouse around measurable business facts and descriptive dimensions, most commonly using Star or Snowflake schemas.

- **Fact & Dimension Tables**

Fact tables hold quantitative measures and foreign keys; **dimension** tables hold descriptive attributes used to filter and group facts.

- **Star Schema**

A central fact table connected directly to denormalised dimension tables, forming a star shape; simple and fast for queries.

- **Snowflake Schema**

An extension of the star schema where dimension tables are normalised into related sub-tables, reducing redundancy but adding joins.

Query Processing

Query processing in a warehouse focuses on answering complex, aggregate analytical queries efficiently using optimisation strategies, specialised indexes and materialised views.

- **Query Optimisation**

OLAP optimisers choose efficient execution plans by exploiting cube aggregates, star-join algorithms and cost-based estimation.

- **Indexing: Bitmap / B-Tree**

Bitmap indexes suit low-cardinality dimension columns for fast AND/OR filtering; B-tree indexes suit high-cardinality range queries.

- **Materialised Views**

Pre-computed, stored query results (aggregate views) speed up repeated analytical queries by avoiding recomputation.

Open-Source BI Tools & KNIME

A range of open-source Business Intelligence tools, including KNIME, provide free, extensible platforms for building ETL pipelines, performing OLAP-style analysis, and generating reports.

- **Open-Source BI Tools**

Examples: Pentaho, Metabase, Apache Superset, Jaspersoft - offering free reporting, dashboards and ETL capabilities.

- **KNIME Analytics Platform**

A visual, node-based, open-source tool where users drag-and-drop nodes to build ETL, analytics and OLAP-style workflows with little coding.

- **KNIME for OLAP Reporting**

KNIME nodes connect to a warehouse and perform group-by / pivot (roll-up, slice, dice equivalents), generating interactive reports and dashboards.

OLAP Application & Cube Computation

This branch covers a hands-on OLAP application built to exercise cube operations, alongside the key algorithms used to pre-compute data cubes: full materialisation, multiway array aggregation, BUC and Star-Cubing.

- **Build an OLAP App**

A sample application (e.g. in KNIME or Python/Pandas) loads a star-schema dataset and performs roll-up, drill-down, slice, dice and pivot interactively.

- **Full Cube Materialisation**

Computes and stores every possible aggregate cuboid in advance for the fastest query response, at the cost of very high storage.

- **Multiway Array Aggregation**

A MOLAP method that computes cube aggregates directly on multidimensional arrays in a single scan, sharing partial sums across cuboids.

- **BUC Algorithm**

Bottom-Up Computation prunes sparse cells using Apriori-style pruning while aggregating bottom-up across dimensions.

- **Star-Cubing**

Combines top-down and bottom-up computation using a star-tree structure to compress data and prune unnecessary aggregation.